

SCHOOL OF ENGINEERING
COCHIN UNIVERSITY OF SCIENCE AND TECHNOLOGY
B. TECH DEGREE 2ND SEMESTER FIRST INTERNAL EXAMINATIONS, APRIL 2024
23-200-0204A: BASIC ELECTRICAL ENGINEERING (CIVIL A BATCH)
(2023 Scheme)

TIME: 2 hrs

MAX. MARKS: 30

Course Outcomes

On completion of this course the student will be able to:

CO1	Explain the concepts of various energy sources and electric circuits
CO2	Apply the basic electrical laws to solve circuits
CO3	Discuss the construction and operation of various electrical machines
CO4	Identify suitable electrical machine for practical implementation

PART A

(Answer all questions)

(5 x 2= 10 Marks)

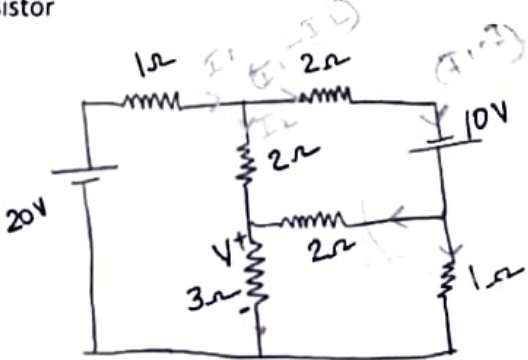
Q.No.	Questions	Marks	BL	CO	PO
I (a)	Why domestic appliances are connected in parallel? Give comparison with series circuit	2	1	2	1
(b)	Three lamps 20W, 60W and 100W connected in parallel across a dc source draw a current of 9.2 A . What is the value of the supply voltage?	2	2	2	1,2
(c)	How does the mutual inductance of a pair of coils change when (i) distance between the coils is increased and (ii) number of turns in the coils is increased	2	1	2	1
(d)	State and explain Lenz's law of electromagnetic induction	2	1	2	1
(e)	State Kirchoff's voltage law. Explain with suitable circuits	2	1	2	1

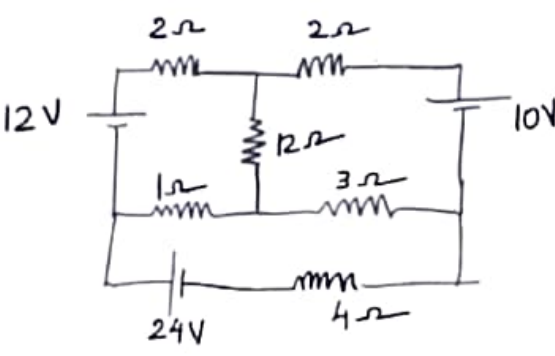
PART B

(Answer the following questions
choosing either the first OR second option)

MODULE 1

(1 x 10= 10 Marks)

Q.No.	Questions	Marks	BL	CO	PO
II (a)	Solve the circuit and find the voltage V across the 3 ohm resistor 	7	3	2	1,2
(b)	State and explain Faradays law	3	1	2	1

OR				
III (a)	Determine the current in 4 ohm branch in the circuit shown by applying Maxwell's loop current method.	7	3	1,2
			2	
(b)	Explain how ohms law is useful in the analysis of a series circuit	3	1	1
MODULE I (1 x 10= 10 Marks)				
IV (a)	With necessary schematic diagram, explain hydroelectric power generation	5	1	1
(b)	The field winding of a dc electromagnet is wound with 960 turns and has resistance of 50ohm. When the exciting voltage is 230V, the magnetic flux linking in the coil is 0.005Wb. Calculate self inductance of the coil	5	2	1,2
OR				
V (a)	General structure of electrical power systems using single line diagram. Explain	5	1	1,2
(b)	A solenoid 70cm in length and of 2100 turns has a radius of 4.5cm. A second coil of 750turns is wound upon the middle of the solenoid. Find mutual inductance of the two coils	5	2	1,2

Bloom's Taxonomy Levels

L1 - %	L2- %	L3-%	L4-%	L5-%	L6-%
48	24	28			